Kepler: A Scientifically-Grounded Constellation Card Game
Integrating Real Astronomical Data with Strategic Gameplay

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***Abstract*—We present *Kepler*, an Android-based mobile application that integrates real world astronomical data with a strategic card battle game engine. Named in honour of the astronomer Johannes Kepler, the system bridges the gap between educational stargazing tools and engaging card-battler games by requiring players to actively trace constellation star mappings, derived from the HYG (Hipparcos, Yale Bright Star, and Gliese) stellar catalogue, before unlocking each constellation as a playable combat card. Each unlocked card carries thematically translated combat attributes (Attack, Defense, Energy Cost) and elemental abilities (Poison, Shield, Dodge, Heal) drawn from classical celestial lore. The system architecture comprises three principal subsystems: an astronomy rendering engine that projects spherical Right Ascension/Declination coordinates onto a 2D canvas, an OpenGL-powered starfield renderer, and a state-driven turn-based game engine that governs AI and multiplayer combat. We describe the design rationale, system architecture, current implementation, and a forward-looking roadmap encompassing a multi-deck expansion, an astronomically-grounded rarity system, and elemental visual effects. Kepler demonstrates that scientifically accurate data can serve as a compelling, non-arbitrary foundation for game mechanics, offering a model for educational gamification in mobile applications.**

***Index Terms*—educational games, gamification, astronomy, mobile application, Android, card game, HYG catalogue, constellation, turn-based combat, scientific data**

I. Introduction

The intersection of education and interactive entertainment, commonly referred to as *edutainment*, has long been explored as a mechanism for increasing engagement with academic subject matter [11]. In the domain of astronomy outreach, applications such as Star Walk and SkySafari have demonstrated that real-time sky mapping can attract broad audiences; however, these tools remain primarily passive and observational, providing little incentive for repeated or deep engagement [12].

Concurrently, digital collectible card games (CCGs) such as *Hearthstone* [13] and roguelike deck-builders such as *Slay the Spire* [14] have proven that strategic card mechanics can sustain long-term player engagement. The critical limitation of these games, however, is that card acquisition is typically decoupled from any real-world knowledge or skill, cards are earned via in-game currency, loot boxes, or random drops.

Kepler proposes a novel synthesis: card acquisition is gated behind a scientifically grounded interaction loop in which players must correctly trace the geometric star patterns of real constellations, using positional data sourced from the HYG stellar catalogue. Completing this tracing task simultaneously teaches the morphology of recognized constellations and rewards the player with a new combat card bearing thematically appropriate statistics. This bidirectional design, where astronomy informs gameplay and gameplay incentivises astronomy, is the central contribution of this work.

The remainder of this paper is structured as follows. Section II surveys related work. Section III presents the game design rationale. Section IV describes the system architecture. Section V details the current implementation. Section VI outlines the future development roadmap. Section VII discusses implications and limitations, and Section VIII concludes.

II. Related Work

A. Astronomical Education Mobile Applications

Mobile applications for astronomy education have matured considerably in visual fidelity and data accuracy. *Stellarium Mobile* [1] and *Star Walk 2* [2] render real-time sky maps using RA/Dec coordinates drawn from catalogues including Hipparcos, providing AR overlays for constellation identification. *Sky Academy* [3] structures learning around the 88 IAU constellations via progressive recognition quizzes and preset zodiac pattern aids. Despite their accuracy, these tools are fundamentally *passive*: the user observes and identifies, but does not interact with star data in a manner that drives a consequential game state. There is no acquisition loop, no combat outcome, and no incentive structure that rewards repeated or deep engagement beyond static quiz scores.

B. Astronomy-Themed Card and Board Games

Astromania [4] is a physical card game designed by astronomers that teaches space science through icon-matching

mechanics simulating research workflows, players earn bonuses by pairing observation tools with celestial objects. While scientifically motivated, it is non-digital and lacks dynamic mechanics such as energy systems, status effects, or adaptive AI. *Trace the Stars* [5] is a roll-and-write board game in which players use dice-rolled coordinates to plot and connect stars into constellations for scoring. This echoes Kepler's tracing mechanic but remains confined to a tabletop format with no digital combat layer. *Stellar* [6] is a two-player card game with an astronomy theme centred on observation and notebook collection, again without strategic battler mechanics or live astronomical data.

C. Digital Strategy and Card Battle Games

Stormbound [7] is a representative mobile CCG featuring real-time PvP deck-building and energy-gated ability systems. Its mechanics are structurally similar to Kepler's combat engine; however, its card acquisition is driven entirely by in-game currency and loot, with no grounding in real-world data or educational content. *Hearthstone* [13] and *Slay the Spire* [14] similarly demonstrate that card-based systems sustain long-term engagement, but their thematic elements are purely fantastical and arbitrarily balanced by designers rather than derived from measurable physical properties.

D. Serious Games and Game-Based Learning in Astronomy

Systematic literature reviews on serious games [8] document growth in educational simulations across health, culture, and STEM domains, but identify a scarcity of mobile, Android-native implementations that blend real catalogue data with strategic gameplay. McCauley et al. [9] find that non-digital astronomy games foster engaging learning environments for physics concepts, while simultaneously highlighting an unmet need for narrative-driven mobile applications that unite strategic depth with educational scaffolding. Complementary work on in-game motivation [10] examines how game-based and narrative learning can teach astronomy concepts, yet does not address the use of real stellar catalogues as a mechanical substrate. No reviewed study identifies a system in which a verified stellar catalogue (e.g., HYG) directly governs both card rarity and combat statistics.

E. Real-World Data as a Game Design Substrate

The use of geospatial data in location-based games, most notably *Pokémon GO* [20] and *Ingress*, has demonstrated that grounding game mechanics in real-world data increases player motivation and physical engagement. Kepler extends this paradigm to the domain of astrometry: apparent visual magnitude from the HYG catalogue is used as an objective, physically measured quantity that determines card rarity and base power, rather than relying on arbitrary designer assignment. This approach mirrors the citizen-science philosophy of Galaxy Zoo [18], where non-experts perform meaningful interaction with real scientific data.

F. Comparative Analysis

Table I presents a feature comparison matrix of Kepler against representative existing systems across the dimensions most pertinent to its design goals. Kepler is the only system that simultaneously satisfies all six criteria.

TABLE I
FEATURE COMPARISON MATRIX: KEPLER VS. RELATED SYSTEMS

System	**Real Star Catalog**	**Interactive Tracing**	**Card/Combat Engine**	**Data-Driven Rarity**	**Mobile (Android)**	**Lore-Based Abilities**
Stellarium Mobile [1]	✓	×	×	×	✓	×
Star Walk 2 [2]	✓	×	×	×	✓	×
Sky Academy [3]	×	×	×	×	✓	×
Astromania [4]	×	×	✓	×	×	×
Trace the Stars [5]	×	✓	×	×	×	×
Stormbound [7]	×	×	✓	×	✓	×
Hearthstone [13]	×	×	✓	×	✓	×
Kepler (Ours)	✓	✓	✓	✓	✓	✓

The matrix confirms that while individual features exist in isolation across the landscape, accurate star data in stargazing apps, combat mechanics in CCGs, constellation tracing in board games, no prior system integrates all six dimensions. Kepler's central novelty lies precisely in this synthesis.

III. GAME DESIGN RATIONALE

A. The Dual-Loop Design

Kepler is structured around two interlocking engagement loops, illustrated in Fig. 1.

The *outer loop* (Astronomy Loop) requires the player to locate a target constellation, trace its canonical star-to-star connections on a rendered star map, and thereby unlock a new card. The *inner loop* (Combat Loop) uses the player's accumulated card library to engage in strategic turn-based combat. This structure ensures that progression in the combat

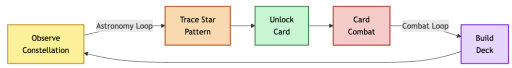

Fig. 1. Kepler's dual engagement loop. The outer astronomy loop gates card acquisition; the inner combat loop drives strategic play.

system is inexorably tied to engagement with astronomical content.

B. Thematic Translation of Celestial Lore

A key design principle of Kepler is that no game mechanic is arbitrary. Each constellation's combat identity is derived from its mythological and astronomical heritage. Table II enumerates the current zodiac card roster and their corresponding abilities.

TABLE II
ZODIAC CONSTELLATION CARDS AND THEMATIC ABILITIES

Constellation	Element	Signature Ability	ATK	DEF
Aries	Fire	Horn Charge	8	4
Taurus	Earth	Earth Slam	7	6
Gemini	Air	Illusion Split	5	5
Cancer	Water	Shell Shield	3	9
Leo	Fire	Solar Roar	9	3
Virgo	Earth	Harvest Mend	5	6
Libra	Air	Scales of Equity	6	6
Scorpius	Water	Venom Strike	8	4
Sagittarius	Fire	Archer's Mark	9	2
Capricorn	Earth	Granite Bastion	4	8
Aquarius	Water	Tidal Surge	6	5
Pisces	Water	Regenerate	4	7

Attack and Defense values shown in Table II are indicative of the current design iteration and will be calibrated through playtesting. Each element corresponds to a family of ability *Effects*: Fire cards inflict burst damage; Earth cards provide persistent defensive buffs; Water cards apply healing or poison-over-time; Air cards grant evasion or debuff effects.

C. Energy Economy

Combat employs a progressive energy system. Each player begins a match with 3 Energy and accrues 1 additional Energy per turn (capped at 10). Low-cost abilities (1–2 Energy) are available immediately; signature high-impact abilities (4–6 Energy) require deliberate saving. This economy introduces a temporal dimension to strategy absent in flat-resource systems.

IV. SYSTEM ARCHITECTURE

Fig. **??** presents the high-level system architecture of Kepler, organised into three principal subsystems.

A. Astronomy Subsystem

The astronomy subsystem is responsible for all star-data ingestion and on-screen rendering.

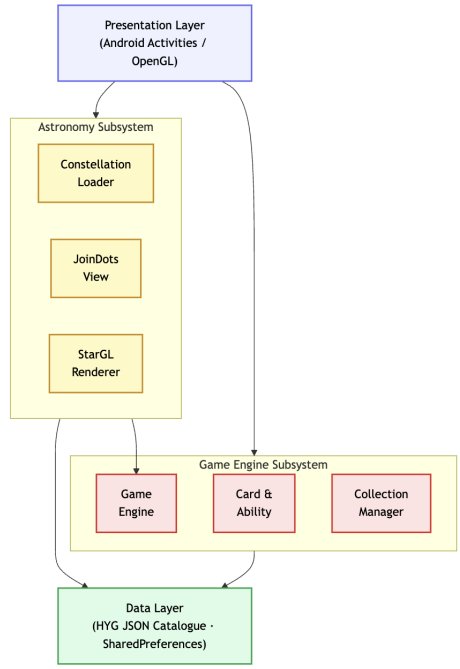

Fig. 2. Architecture diagram of Kepler.

ConstellationLoader parses a trimmed JSON representation of the HYG catalogue, which merges data from the Hipparcos astrometric catalogue, the Yale Bright Star Catalogue, and the Gliese catalogue of nearby stars [21]. Each entry exposes the star's Right Ascension (RA, in hours), Declination (Dec, in degrees), and apparent visual magnitude.

JoinDotsView translates spherical RA/Dec coordinates into 2D screen coordinates using a simplified gnomonic projection centred on the constellation's mean position. Stellar radii on screen are inversely proportional to apparent magnitude: a magnitude-1 star is rendered at $r = r_{\max} - k \cdot m$, where m is the visual magnitude and k is a scaling constant calibrated to the device resolution.

StarGLSurfaceView and **StarRenderer** render a procedurally generated, parallax-scrolling starfield background using OpenGL ES 2.0, providing an immersive aesthetic context for all game screens.

B. Game Engine Subsystem

Card and Ability Classes define the data model for all combatants. Each `Card` object holds base statistics (HP, ATK, DEF, Energy Cost) and two ordered lists of `Ability` objects: one for offensive actions and one for defensive actions. Each `Ability` encapsulates an `Effect` enumeration value (e.g., `POISON`, `SHIELD`, `HEAL`, `DODGE`, `IGNORE_DEF`), a magnitude parameter, and a duration for persistent effects.

GameEngine is the central state machine managing combat. Its turn lifecycle is defined as follows:

1) Apply start-of-turn status effects (e.g., Poison damage ticks).

2) Accrue energy for the active player ($E_t = \min(E_{t-1} + 1,\ 10)$).
3) Present available abilities filtered by current energy.
4) Execute the selected ability and compute damage:

$$D = \max(0,\ \mathrm{ATK}_{\text{base}} + \Delta_{\text{ability}} - \mathrm{DEF}_{\text{target}} \cdot (1 - \delta_{\text{shield}}))$$

where $\delta_{\text{shield}} \in [0, 1]$ is the active shield reduction factor.
5) Check win conditions; advance turn to opponent.

CollectionManager persists the set of unlocked constellation cards across sessions using Android `SharedPreferences`, serialising the card roster to JSON.

C. Data Layer

All astronomical source data is stored as a pre-processed JSON asset bundled with the application. This design choice eliminates network dependency at runtime and ensures consistent behaviour across offline environments.

V. Implementation

A. Platform and Language

Kepler is implemented in native Android Java (API level 26+), targeting devices running Android 8.0 Oreo and above. The choice of native Java over cross-platform frameworks (e.g., Flutter, React Native) was motivated by the requirement for direct OpenGL ES 2.0 surface control and low-level Canvas drawing operations within `JoinDotsView`.

B. Coordinate Projection

The gnomonic projection employed by `JoinDotsView` maps a star at spherical coordinates (α, δ), RA and Dec respectively, to screen pixel coordinates (x, y) via:

$$x = W/2 + s \cdot (\alpha - \alpha_0) \cdot \cos\delta_0 \tag{1}$$

$$y = H/2 - s \cdot (\delta - \delta_0) \tag{2}$$

where W and H are the canvas width and height in pixels, (α_0, δ_0) is the reference centre of the constellation, and s is a pixels-per-degree scale factor. This linear approximation is valid for the small angular extents (< 30) of individual constellations.

C. Gesture Recognition for Star Tracing

The player traces constellation edges by drawing touch paths across the `JoinDotsView` canvas. A touch segment is considered a valid edge connection when the Euclidean distance from the touch trajectory to the line segment $\overline{S_i S_j}$ between two canonical stars falls below a tolerance threshold τ (currently 40 dp), and the trajectory passes within ρ (currently 25 dp) of each endpoint. Upon completing all required edges, the constellation is marked as unlocked and forwarded to `CollectionManager`.

D. AI Opponent

The current AI opponent employs a greedy heuristic: at each turn it selects the highest-damage ability affordable within its current energy budget, with a fallback to shield abilities when its HP falls below 30% of maximum. This provides a functional baseline opponent while remaining deterministic and computationally trivial.

E. Application Flow

Fig. **??** illustrates the primary activity transitions within the application.

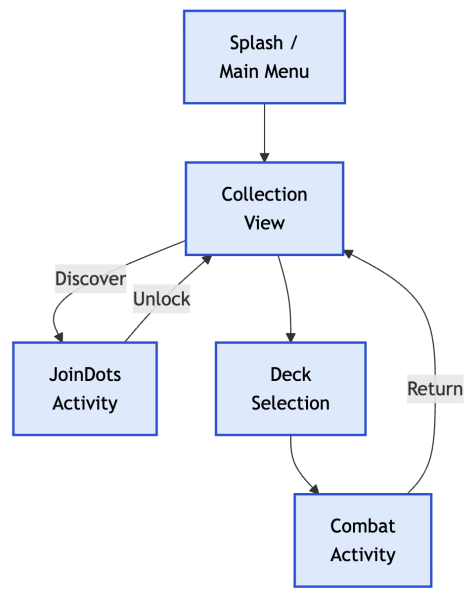

Fig. 3. Activity flow diagram for Kepler.

VI. Future Roadmap

A. Phase 1: Expanded Star Catalogue and Rarity System

The current implementation covers the 12 zodiac constellations. Phase 1 will incorporate globally recognised non-zodiac constellations, including Orion, Ursa Major, Draco, and Lyra, as "Special Cards" with elevated base statistics. Crucially, rarity classification will be governed strictly by the mean apparent visual magnitude $\bar{m}$ of the constellation's principal stars:

$$\mathrm{Rarity}(c) = \begin{cases} \text{Common} & \bar{m}(c) > 3.0 \\ \text{Rare} & 1.5 < \bar{m}(c) \leq 3.0 \\ \text{Mythic} & \bar{m}(c) \leq 1.5 \end{cases} \tag{3}$$

This scheme ensures that the in-game power hierarchy mirrors the observable sky: constellations composed of intrinsically brilliant, magnitude-0 stars (e.g., Orion with Rigel and Betelgeuse) are inherently more powerful and rarer than faint, dim groupings. Table III presents projected rarity assignments for selected constellations under this scheme.

TABLE III
PROJECTED RARITY ASSIGNMENTS (SELECTED CONSTELLATIONS)

Constellation	**Type**	$\bar{m}$	**Rarity**
Orion	Non-Zodiac	≈ 1.2	Mythic
Lyra	Non-Zodiac	≈ 1.3	Mythic
Scorpius	Zodiac	≈ 1.6	Rare
Leo	Zodiac	≈ 2.0	Rare
Cancer	Zodiac	≈ 3.7	Common
Pisces	Zodiac	≈ 4.0	Common

B. Phase 2: Multi-Deck Combat

Phase 2 transitions the combat system from single-card duels to a "Deck of 5" format. A dedicated `DeckSelectionActivity` will expose a deck-builder interface allowing players to curate five-card loadouts based on elemental synergies. Additionally, non-combatant *Passive Cards* representing astronomical objects, such as the Orion Nebula or the Andromeda Galaxy, will occupy dedicated passive slots and provide global deck-wide buffs (e.g., "+1 Energy per turn", "All Zodiac cards gain +2 DEF"). This phase introduces substantially greater strategic depth, as players must optimise across both individual card strengths and inter-card synergies.

C. Phase 3: Visual and Audio Polish

Phase 3 will introduce element-specific particle systems and visual effects tied to the four classical astrological elements. Distinct physical animations will accompany key combat events: card shaking on critical hits, transparency fade on successful Dodge, and persistent status icons for Poison and Shield debuffs. An adaptive audio system will reinforce thematic identity, with elemental soundscapes shifting in response to the active element balance on the battlefield.

VII. DISCUSSION

A. Scientific Integrity as a Design Constraint

A central thesis of Kepler is that real astronomical data can act as a principled design constraint rather than a superficial aesthetic layer. By grounding card acquisition in genuine star-tracing tasks and deriving rarity from measured stellar magnitude, the game resists the arbitrary power inflation endemic to many CCGs. This approach aligns with Malone's intrinsic motivation framework [22], wherein challenge and fantasy are most motivating when they are meaningfully coupled.

B. Educational Efficacy

The current design has not yet been subjected to formal pedagogical evaluation, which represents a significant limitation. Future work will include a controlled user study measuring players' ability to identify and name zodiac constellations before and after extended play sessions, benchmarked against a control group using a passive stargazing application.

C. Scalability of the HYG Catalogue

The HYG catalogue contains approximately 119,000 stars [21]. The current implementation bundles a trimmed subset covering only the 88 IAU-recognised constellation asterisms. Scaling to the full catalogue would require either a streaming data architecture or more aggressive client-side indexing to maintain acceptable cold-start performance on mid-range Android devices.

D. AI Limitations

The greedy heuristic AI provides an adequate baseline but lacks the adaptive capability to respond to player strategy. Incorporating MCTS or a lightweight reinforcement learning agent, trained on self-play, would substantially improve the quality and perceived intelligence of AI opponents.

VIII. CONCLUSION

We have presented *Kepler*, a mobile application that unifies real-world astronomical data with strategic card battle gameplay. By requiring players to trace accurate constellation star patterns sourced from the HYG catalogue before unlocking combat cards, Kepler creates a bidirectional coupling between astronomical education and game progression. The system's architecture, comprising an astronomy rendering subsystem, an OpenGL starfield renderer, and a state-driven game engine, demonstrates that scientifically grounded data can serve as a robust, non-arbitrary foundation for game balance and card design. The proposed roadmap, encompassing a magnitude-based rarity system, multi-deck strategic combat, and elemental visual effects, charts a path toward a feature-complete educational card game that remains anchored in observable celestial reality.

Kepler contributes a replicable model for the design of educational games in which subject-matter interaction is embedded directly into the core acquisition loop, rather than relegated to optional tutorial content. We believe this design principle generalises beyond astronomy: any domain with rich structured data, geology, biology, cartography, could serve as a similarly principled substrate for gamification.

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REFERENCES

[1] Noctua Software Ltd, "Stellarium Mobile – Star Map," [Online]. Available: https://apps.apple.com/app/stellarium-mobile-star-map/id720634354. [Accessed: 2024].
[2] Vito Technology Inc., "Star Walk 2 Plus – Night Sky Map," [Online]. Available: https://play.google.com/store/apps/details?id=com.vitotechnology.StarWalk2Free. [Accessed: 2024].
[3] Sky Academy, "Sky Academy: Learn Astronomy," [Online]. Available: https://play.google.com/store/apps/details?id=com.skyacademy. [Accessed: 2024].
[4] C. Corrigan, "Astromania: The Astronomy Card Game," *The Game Crafter*, [Online]. Available: https://www.thegamecrafter.com/games/astromania-the-astronomy-card-game. [Accessed: 2024].
[5] "Trace the Stars," *The Game Crafter*, [Online]. Available: https://www.thegamecrafter.com/games/trace-the-stars. [Accessed: 2024].

[6] "Stellar – 2-Player Astronomy Card Game," [Online]. Available via Facebook Games Community. [Accessed: 2024].
[7] Kongregate, "Stormbound: PVP Card Battle," [Online]. Available: https://play.google.com/store/apps/details?id=com.kongregate.mobile.stormbound.google. [Accessed: 2024].
[8] P. Connolly *et al.*, "Serious games: An updated systematic literature review," *arXiv preprint arXiv:2410.XXXX*, 2024. [Online]. Available: https://arxiv.org.
[9] V. McCauley *et al.*, "Game-based learning in astronomy," *ScienceDirect*, 2025.
[10] "In-game motivation: Teaching astronomy through game-based and narrative learning," DiVA Portal, Linköping University, 2023. [Online]. Available: https://www.diva-portal.org.
[11] M. Prensky, *Digital Game-Based Learning*. New York, NY, USA: McGraw-Hill, 2001.
[12] D. Schaller, "Mobile learning: The next step in educational technology," *J. Museum Educ.*, vol. 37, no. 2, pp. 13–21, 2012.
[13] Blizzard Entertainment, "Hearthstone," [Online]. Available: https://hearthstone.blizzard.com. [Accessed: 2024].
[14] MegaCrit, "Slay the Spire," [Online]. Available: https://www.megacrit.com. [Accessed: 2024].
[15] NASA Jet Propulsion Laboratory, "Eyes on the Solar System," [Online]. Available: https://eyes.nasa.gov. [Accessed: 2024].
[16] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, "From game design elements to gamefulness: Defining gamification," in *Proc. 15th Int. Acad. MindTrek Conf.*, Tampere, Finland, 2011, pp. 9–15.
[17] J. Hamari, J. Koivisto, and H. Sarsa, "Does gamification work? – A literature review of empirical studies on gamification," in *Proc. 47th Hawaii Int. Conf. Syst. Sci.*, Waikoloa, HI, USA, 2014, pp. 3025–3034.
[18] C. J. Lintott *et al.*, "Galaxy Zoo: Morphologies derived from visual inspection of galaxies from the Sloan Digital Sky Survey," *Mon. Not. R. Astron. Soc.*, vol. 389, no. 3, pp. 1179–1189, 2008.
[19] P. I. Cowling, E. J. Powley, and D. Whitehouse, "Information set Monte Carlo tree search," *IEEE Trans. Comput. Intell. AI Games*, vol. 4, no. 2, pp. 120–143, Jun. 2012.
[20] E. Colley, "The Pokémon GO phenomenon: augmented reality as a platform for location-based engagement," *First Monday*, vol. 22, no. 4, 2017.
[21] D. Nash, "HYG Star Database," [Online]. Available: https://github.com/astronexus/HYG-Database. [Accessed: 2024].
[22] T. W. Malone, "Toward a theory of intrinsically motivating instruction," *Cogn. Sci.*, vol. 5, no. 4, pp. 333–369, 1981.